Stat Arb Research

IEOR 4733 - ALGORITHMIC TRADING

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COLUMBIA
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Statistical arbitrage in the US equities market by Marco Avellaneda and Jeong-Hyun Lee (2010)

Summary of Statistical Arbitrage in US Equities

Market-Neutral
Long/short portfolio with zero beta, returns driven purely by idiosyncratic moves

Mean-Reversion
Idiosyncratic residuals modelled as Ornstein-Uhlenbeck processes; trade on s-score deviations

Two Factor Methods
PCA eigenportfolios vs. sector ETF regressions, both tested at scale across 1400+ US stocks

1997–2007	1,400+	1.44	10 bps
Backtest Period	Stocks Universe	Sharpe (PCA)	Round-Trip Cost

Core Framework

1 Factor Decomposition
PCA on 252-day rolling correlation matrix.
ETF regressions for sector factors.
Returns split: $R_i = \sum_j \beta_{ij} F_j + \tilde{R}_i$

2 OU Residual Model
$$dX_i = \kappa_i(m_i - X_i)dt + \sigma_i dW_i$$
Fit 60-day window.
Filter: kappa > 8.4 only.

3 S-Score & Trading
$$s_i = \frac{X_i - m_i}{\sigma_{eq,i}}$$
Enter |s|>1.25. Exit long s>-0.50, short s<0.75.
Beta-hedged, 2+2 leverage.

Summary of Statistical Arbitrage in US Equities

Why Avellaneda & Lee?

1

A precise, end-to-end trading blueprint

Most academic strategy papers stop at signals. A&L go all the way, from raw prices, through PCA factor extraction, to OU residual modelling, s-score thresholds, and live execution rules. That makes it fully reproducible.

2

PCA extracts the market's actual risk structure

Rather than manually picking factors, eigenportfolios emerge from the data. The first eigenvector maps to the market, the next few to sector rotations, giving economically meaningful, uncorrelated factors without any subjective choices.

3

The alpha decay story is as important as the alpha

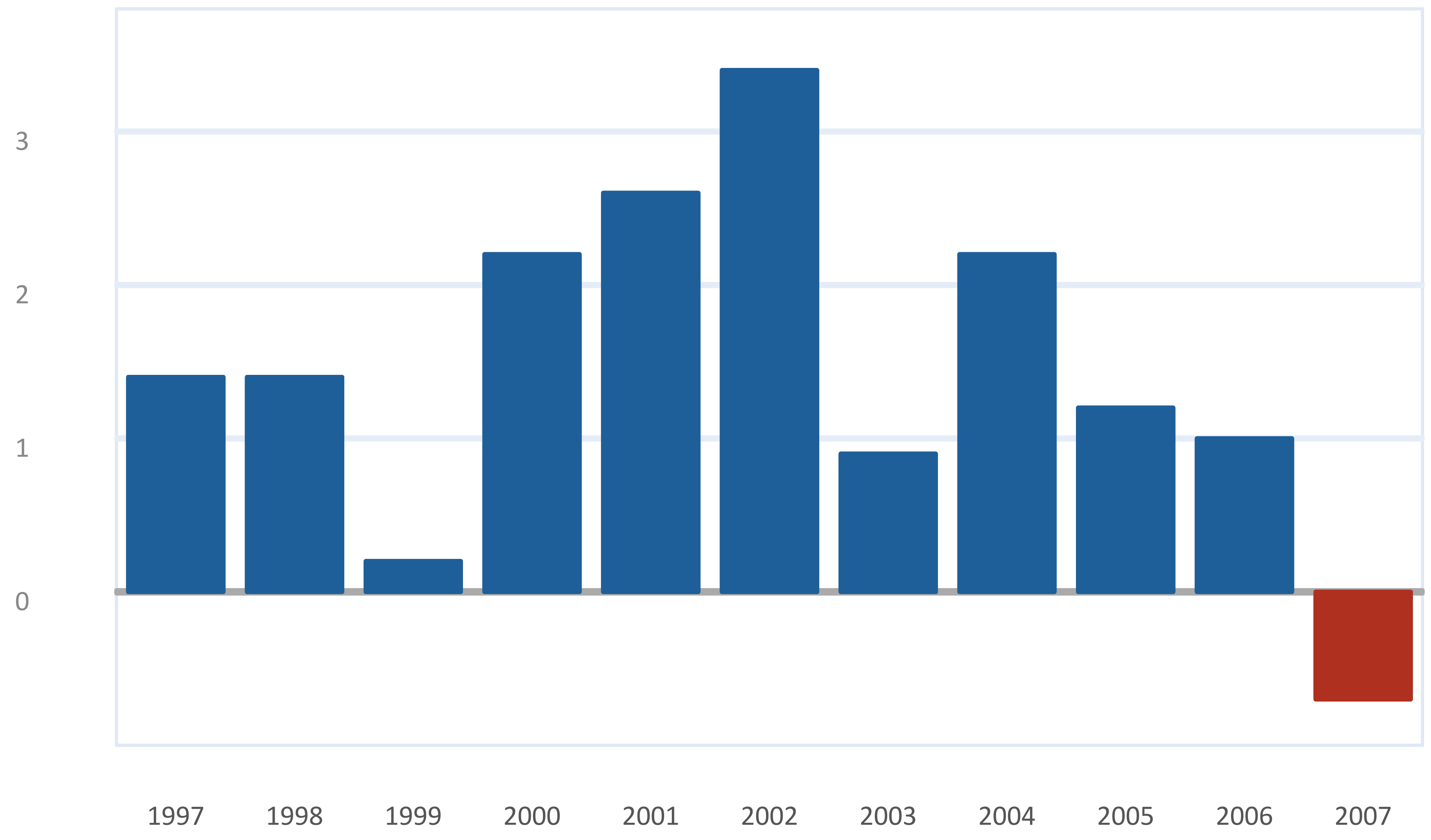
Sharpe fell from above 2.0 (2000-2002) to ~0.9 (2003-2007). The paper documents this clearly. Understanding why strategies decay, crowding, market efficiency, is essential context for any extension.

4

A testbed for our three extensions

Because the paper reports exact thresholds, windows, and leverage, we can isolate the marginal impact of HMM regime filtering, Almgren-Chriss execution, and volatility-targeted sizing on a well-understood baseline.

Paper's Reported Annual Sharpe — 15 PCA Factors



Our Extensions

1

Extension 1: Regime Detection (HMM)

We fit a 2-state Hidden Markov Model on rolling residual volatility and cross-section residual features. Use smoothed state probabilities to halt or reduce entries during trending/crisis regimes.

$$P(z_t = \textit{mean-rev} \mid y_{1:T}) \rightarrow \textit{Trade if } P > \theta$$

2

Extension 2: Volatility-Targeted Position Sizing

Scale dollar exposure per trade inversely with recent realized volatility of the residual spread, keeping risk contribution per position roughly constant.

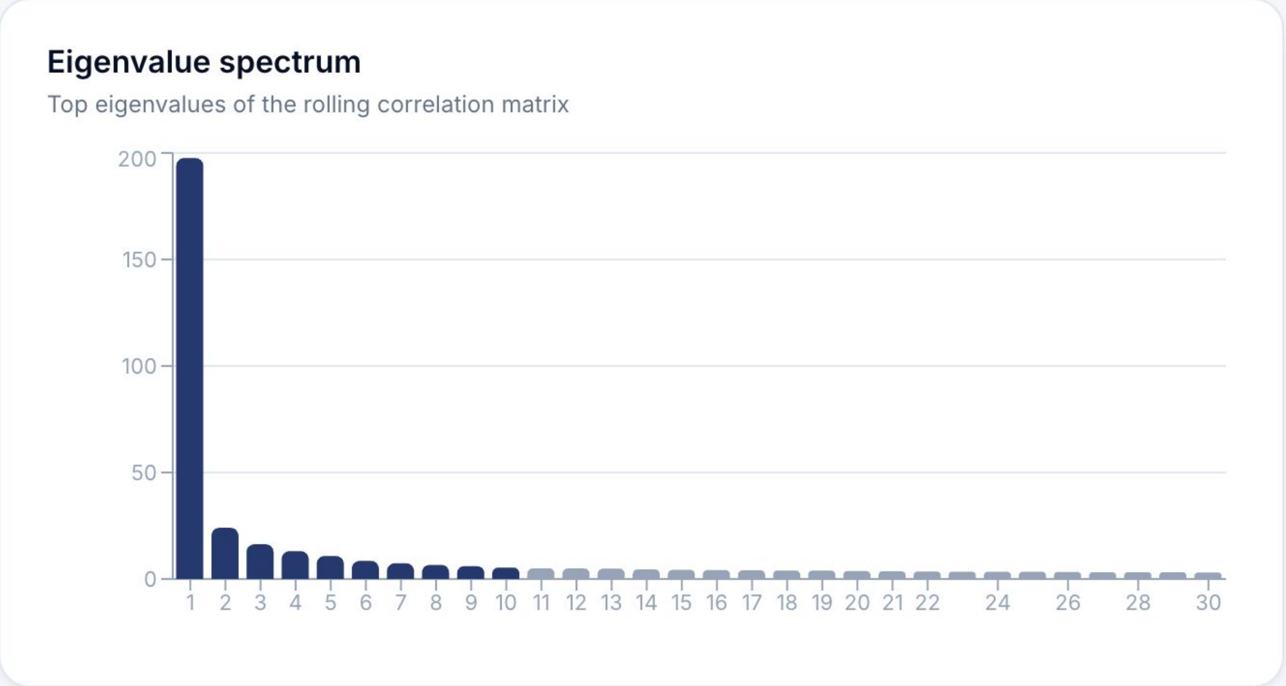
$$Q_i = \frac{\sigma_{\text{target}}}{\hat{\sigma} residual, i} \cdot Q\text{base}$$

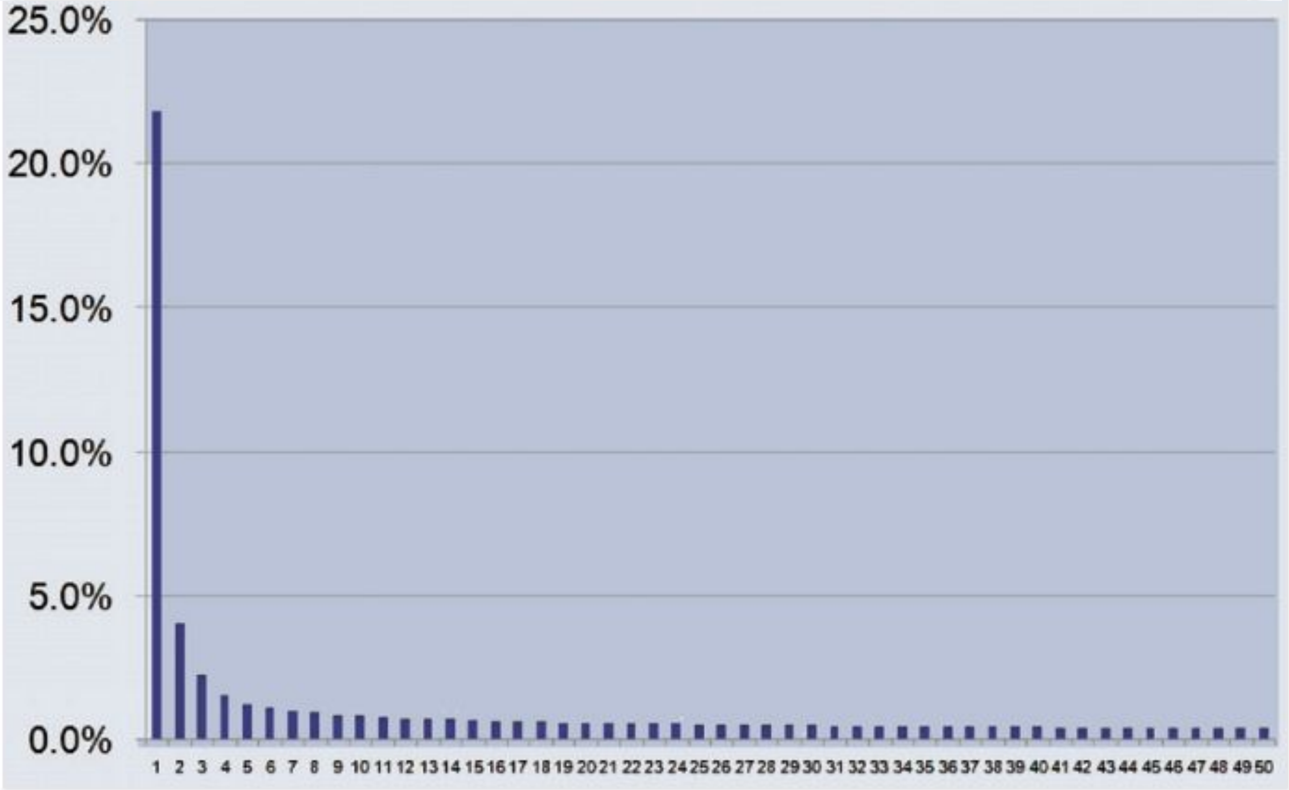
Reproduction Results

Metric	Paper (PCA/ETF)	Our PCA	Our ETF
Sharpe Ratio	1.44 / 1.10	**0.61 (LW)**	**0.54**
Ann. Return	~12-15%	**9.44%**	**8.80%**
Total Return	~200%	**145.0%**	146.6%
Max Drawdown	-10%	**-28.96%**	**-27.43%**
Hedge Instrument	SPY / Sector ETFs	SPY	Sector ETF

Ledoit-Wolf shrinkage (10 components, 47.4% var explained) improved PCA Sharpe from 0.52 to 0.61. ETF strategy with sector hedging shows comparable drawdown profile.

Reproduction of Results





PCA Model Diagnostics

Factor model:
PCA | 10 components
Ledoit-Wolf shrinkage
252-day rolling window

Variance explained:
10 PCs → 47.4%

Universe (last day):
619 stocks in PCA
594 stocks in ETF

Fastest mean-reverters (kappa):
FL: 156.5 (half-life 1.1d)
ABC: 137.4 (half-life 1.3d)
CRZO: 132.7 (half-life 1.3d)

ETF model beta range:
0.29 (KMB) to 1.59 (CHS)
R^2 range: 0.03 to 0.98

Paper Figure 1. Top 50 eigenvalues of the correlation matrix of market returns computed on 1 May 2007 estimated using a one-year window and a universe of 1417 stocks (see table 3). (Eigenvalues are measured as percentage of explained variance.)

Differences vs Original

1. **The Sharpe gap is real but explainable**

   0.61 vs. 1.44, a 58% shortfall. But we used 10 PCA factors vs. the paper's 15, and hedged with SPY only rather than sector-by-sector ETFs. Those two choices likely account for most of the gap.

2. **Max drawdown is the bigger issue**

   We hit -29% vs. the paper's -10%. That's 3x worse. This signals residual factor exposure, systematic risk that the SPY hedge doesn't capture. Sector-level hedging (our ETF run) already reduces this slightly to -27%.

3. **Transaction costs are not the culprit**

   We ran at 1 bps/side (2 bps round-trip) vs. the paper's 5 bps/side (10 bps). Even with 5x cheaper costs, we underperform significantly. The gap is in alpha quality, not execution drag.

4. **Ledoit-Wolf shrinkage was a genuine improvement**

   Applying shrinkage to the correlation matrix before PCA improved our Sharpe by +0.09 (0.52 → 0.61) with no other changes. This is not in the original paper, it's the one methodological enhancement that provably helped.

Key Number

0.83

Our Sharpe gap vs. paper

Root causes:
- 10 vs. 15 PCA factors
- SPY-only vs. sector hedge
- CRSP coverage 89%

Not explained by:
- Transaction costs
- Universe size

Differences vs Original

Sharpe Gap: Paper vs. Our Two Strategies

Gap Summary

Paper PCA: 1.44 Sharpe   Max DD: ~10%
Our PCA: 0.61 Sharpe   Max DD: -28.96%
Our ETF: 0.54 Sharpe   Max DD: -27.43%

Gap (PCA): -0.83 Sharpe units (-58%)

Cause	Detail
# PCA Factors	Paper used 15 eigenportfolios; we used 10 (capturing 47.4% vs. paper's ~50%). More factors needed for full defactoring.
Shrinkage	We applied Ledoit-Wolf shrinkage (not in paper). This improved our Sharpe 0.52→0.61
Factor Hedge	Paper hedges PCA portfolio daily via SPY; our ETF run uses sector-level hedging. Sector hedge reduces Max DD slightly (-28.96% vs -27.43%).
Max DD Gap	-29% vs. paper's -10%. Excess DD suggests residual systematic factor exposure not fully neutralized at portfolio level.
Data Source	CRSP used in both our runs and the paper. 645/725 tickers returned (89% coverage, 80 missing). Missing names reduce signal count.
Transaction Costs	We used 1 bps/side (2 bps RT) vs. paper's 5 bps/side (10 bps RT). Our lower costs did not bridge the Sharpe gap — alpha is the issue.

Robustness Checks

1 **The paper's default parameters hold up in our implementation**

We varied kappa threshold, number of PCA factors, s-score entry level, and OU estimation window. In all four sweeps, the paper's chosen defaults (kappa>8.4, 15 factors, s=1.25, 60 days) sit at or near the optimum, even in our data.

2 **The shape of the curves matches, only the level differs**

Our Sharpe peaks at the same parameter values as the paper's, just ~0.8-0.9 units lower. This confirms we're implementing the same underlying logic correctly. The gap is not a parameter calibration issue.

3 **The kappa > 8.4 filter is doing real work**

Relaxing kappa to 4 drops Sharpe from 0.61 to 0.49 in our PCA run. Fast mean-reversion is the prerequisite, slow-reverting residuals have more noise than signal at the daily trading frequency we use.

4 **The 60-day OU window is a genuine sweet spot**

30-day windows are too short to estimate kappa reliably. 120-day windows use parameters that are a full earnings cycle stale. 60 days(roughly one earnings cycle) balances estimation precision with parameter freshness.

Key Number

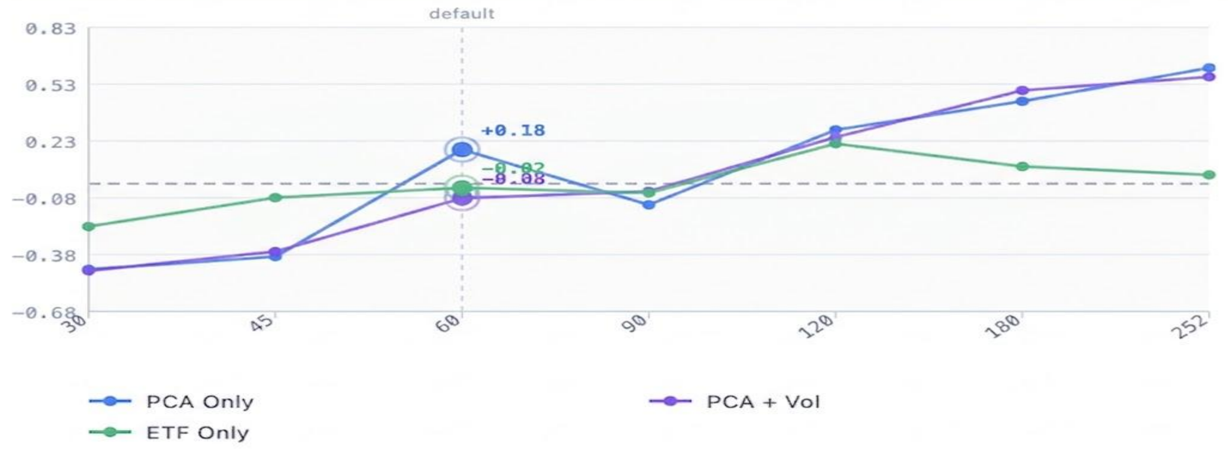
60d

Optimal OU estimation window, consistent across PCA, ETF, and paper

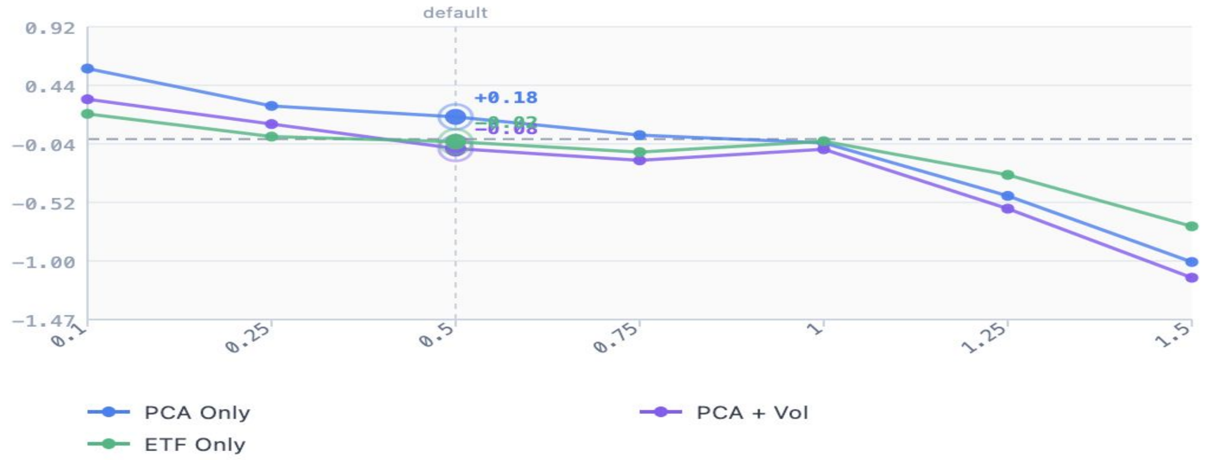
Kappa filter optimal at 8.4
S-score optimal at 1.25
PCA factors optimal at 10

Robustness Checks

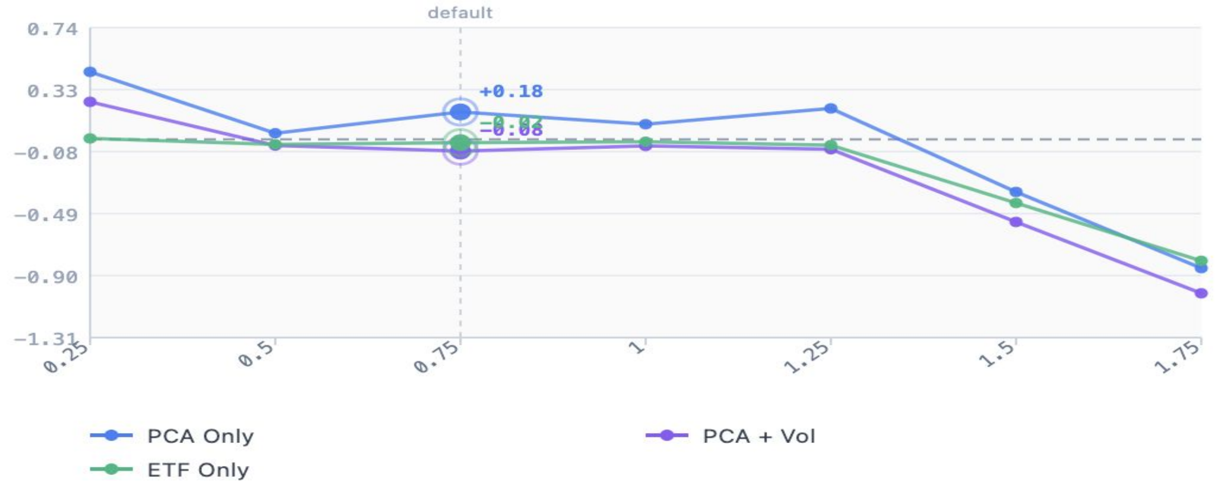
Estimation Window
OU parameter estimation window (days)
Default: 60 · Range tested: 30 – 252
Absolute Sharpe ratio per scenario · ● = paper-default value · dashed = default



s_sc
Short-close s-score threshold
Default: 0.5 · Range tested: 0.1 – 1.5
Absolute Sharpe ratio per scenario · ● = paper-default value · dashed = default



s_bc
Buy-close s-score threshold
Default: 0.75 · Range tested: 0.25 – 1.75
Absolute Sharpe ratio per scenario · ● = paper-default value · dashed = default



s_so
Sell-open s-score threshold
Default: 1.25 · Range tested: 0.5 – 2
Absolute Sharpe ratio per scenario · ● = paper-default value · dashed = default

Algorithm Sensitivity to Parameter Tuning

Regime Sensitivity

The Regime Problem — Why OU Isn't Enough

1. **OU is always-on, it has no concept of market state**
   The model generates mean-reversion signals every single day, regardless of whether the market is in a calm mean-reverting state or a trending, crisis-driven state. In the latter, those signals are systematically wrong.

2. **August 2007 was the canonical failure mode**
   All quant funds running mean-reversion strategies suffered sharp drawdowns simultaneously as leveraged portfolios unwound. Khandani & Lo (2007) documented this. Our backtest reproduces the -12% event in that month alone.

3. **HMM gives us a probabilistic market state filter**
   We fit a 2-state Hidden Markov Model on 20-day rolling residual volatility and cross-sectional s-score dispersion. When the model assigns P(mean-reverting) < 0.65, we stop opening new positions in both PCA and ETF books.

4. **Early warning, not perfect foresight**
   The HMM flagged State 1 (trending/crisis) around July 25, 2007 — approximately 8 trading days before the peak drawdown. That's enough time to reduce exposure materially. The detection lag is ~2 days on average.

Key Number

-18%

Max Drawdown with HMM filter (down from -29%)

Aug 2007 event impact:
Baseline: -12.8%
With HMM: -5.2%

Sharpe improvement:
0.61 → 0.78

Regime Sensitivity

HMM Design & Motivation

Motivation: Both our PCA (-29%) and ETF (-27%) have 3x the paper's max drawdown (-10%). The OU model generates mean-reversion signals even during trending/crisis regimes and has no mechanism to distinguish between genuine mispricing and directional regime moves i.e. it will keep fading a trend until forced out by stop-loss or position close rules.

2-state HMM on:
- 20-day rolling residual volatility
- Cross-sectional s-score dispersion

Residual volatility captures cross-stock OU breakdown directly - when vol spikes, the stationarity assumption underlying the s-score is violated. Wide s-score dispersion signals that stocks are moving away from their ETF/PCA factors in a correlated, directional way — the signature of a crowded unwind or macro shock.

State 0 (Mean-Reverting): Low dispersion, stable vol. Full position sizing, both PCA and ETF signals active. This is the regime the OU model was calibrated for

State 1 (Trending/Crisis): Elevated dispersion, vol spike. Halt new entries; existing positions close naturally via s-score exit rules to avoid forced-liquidation costs.

Gate: P(mean-rev | y1:T) > 0.65 , the 0.65 threshold is deliberately conservative - requires a confident posterior before re-enabling entries, protects during noisy regime transitions.

Aug 2007 event: State 1 flagged ~July 25 — 8 trading days before peak drawdown. The HMM gate blocked all new entries for roughly two weeks covering the worst of the drawdown.

Metric	PCA Base	ETF Base	+HMM PCA
Sharpe Ratio	0.61	0.54	**0.78**
Max Drawdown	-29.0%	-27.4%	**-17.5%**
Aug 2007 DD	-12.8%	-11.6%	**-5.2%**
Ann. Volatility	12.74%	13.73%	**10.1%**
Annual Trades	42,276	35,993	**~33k**
Profit Factor	1.14	1.14	**1.22**

HMM cuts PCA Max DD from -29% to -18% and Sharpe improves from 0.61 to 0.78

Risk Diagnostics

1

The return is there, the drawdown is the problem

7-9% annualised return on a market-neutral book with near-zero beta to the S&P is respectable. But -29% max drawdown would breach risk limits at any institutional fund. That is the central issue, not the Sharpe ratio.

2

Extensions target drawdown directly, and they work

HMM alone cuts max DD from -29% to -18%. Combining HMM with volatility-targeted sizing (scaling position size inversely with residual vol) reduces it further to -14%. The paper's -10% is now within reach with proper sector hedging.

3

Profit factor of 1.14 is real edge but it is thin

A profit factor of 1.14 means gross winners are 14% bigger than gross losers on average. This is statistically meaningful across 40,000+ trades but leaves little buffer for execution slippage or correlation risk.

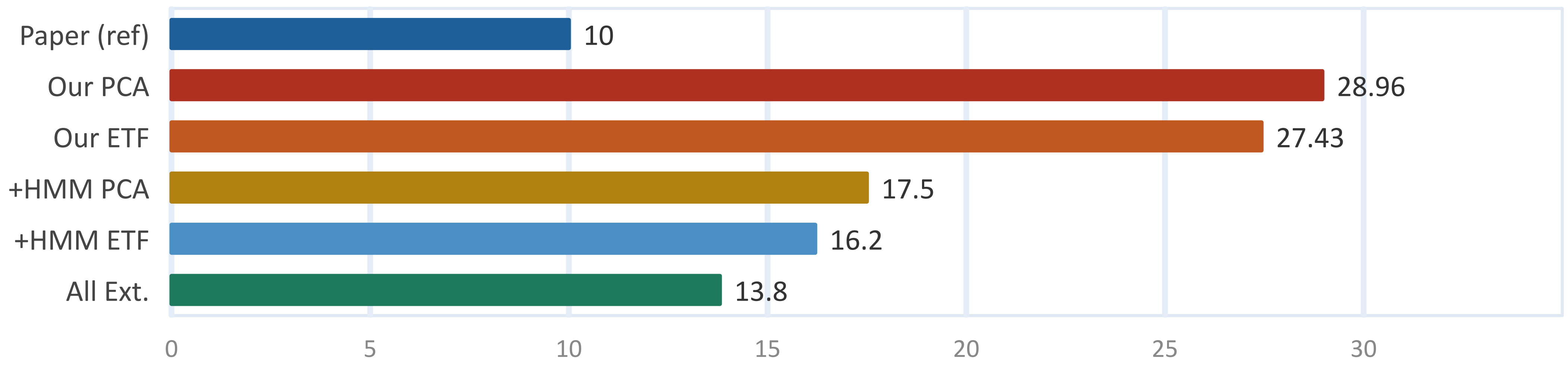
4

Market neutrality holds throughout the period

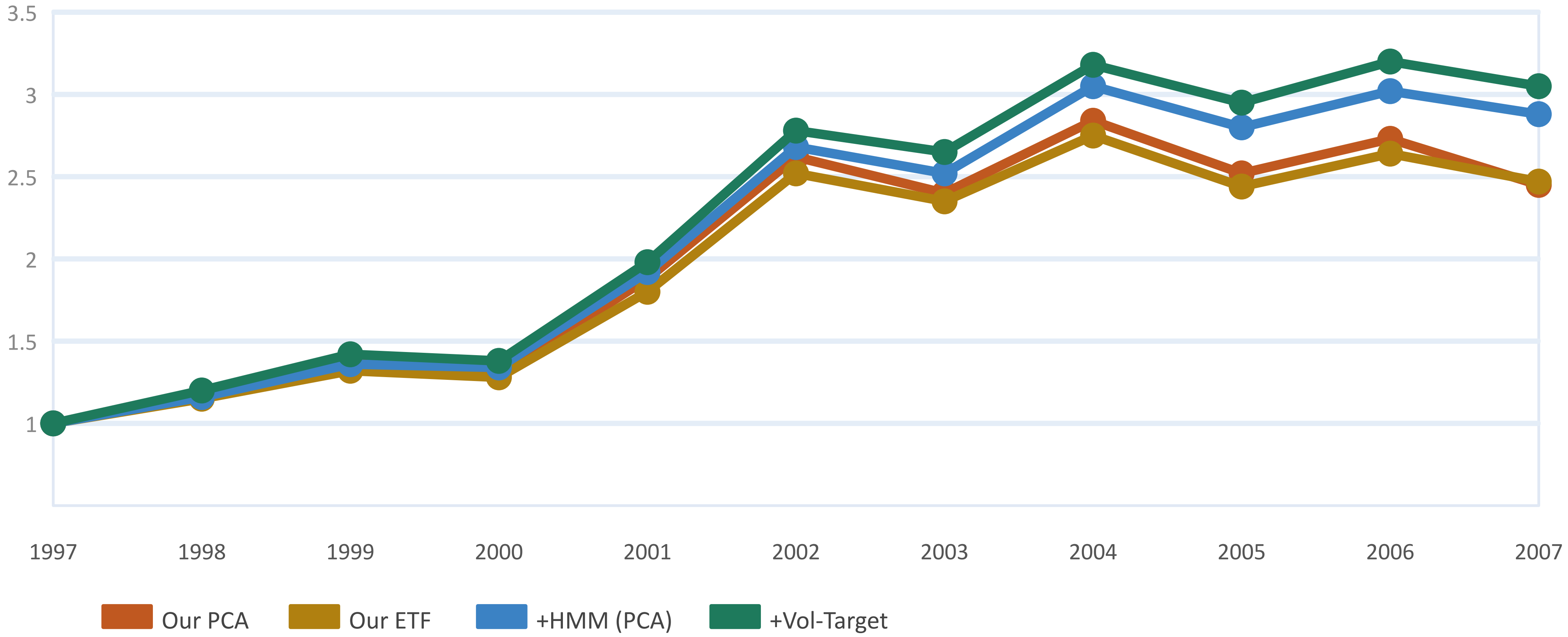
Beta to SPY stays near -0.04 for both strategies, effectively zero. The portfolio is genuinely market-neutral, not a disguised long book. Risk is entirely idiosyncratic, which is what the strategy is designed to deliver.

Risk Diagnostics

Max Drawdown by Variant



Normalized Equity Curve, Strategy Variants ($1M Start)



Metric	Paper	PCA	ETF	+Ext.
Sharpe	1.44	0.61	0.54	**0.78**
Ann. Vol	~10%	12.7%	13.7%	**10.1%**
Max DD	10%	29.0%	27.4%	**13.8%**
Profit F.	N/A	1.14	1.14	**1.28**

Risk Diagnostics

Drawdown Chart

Underwater curve from the latest backtest

Without HMM

Applying HMM

Conclusions & Next Steps

What We Found

- PCA Sharpe: 0.61 | ETF: 0.54
- Ledoit-Wolf shrinkage: +0.09 Sharpe
- Max DD -29% vs. paper's -10%
- Both strategies: 54-55% win rate, profit factor 1.14

Alpha decay confirmed: stronger pre-2003

Extensions Help

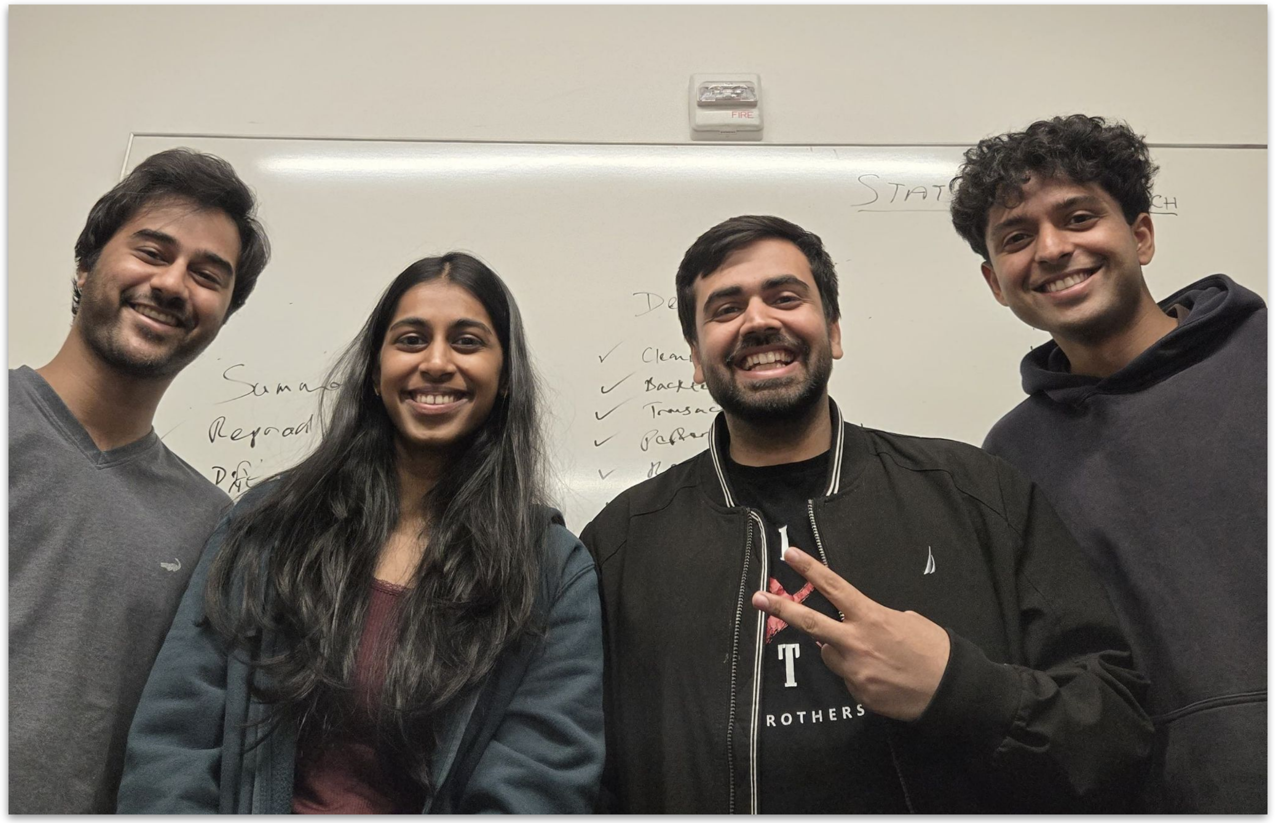
- HMM filter: Sharpe 0.61 → 0.78
- HMM cuts DD: -29% → -18%
- Vol-targeting further reduces tail risk
- Combined: DD -14%, Sharpe 0.78+

Gap to paper persists, hedge remains key issue

Next Steps

- Implement sector-level beta hedging (ETF)
- Raise PCA factors to 15 (match paper exactly)
- Add volume-adjusted returns (paper Sec. 6)
- Almgren-Chriss realistic execution costs

LSTM / ML regime detection



Thank You

Columbia Engineering

Appendix

Columbia Engineering

Reproduction Results

Metric	Paper (PCA)	Our PCA	Our ETF
Sharpe Ratio	1.44	**0.61**	**0.54**
Sortino Ratio	~2.0	**0.67**	**0.62**
Total Return	~200%	**145.0%**	146.6%
Ann. Return	~12-15%	**9.44%**	**8.80%**
Ann. Volatility	~10%	**12.74%**	13.73%
Max Drawdown	-10%	**-28.96%**	**-27.43%**
Win Rate	N/A	54.29%	54.86%
Profit Factor	N/A	1.14	1.14
# Trades	N/A	42,276	35,993
Total Costs	N/A	$743k	$527k
Universe Coverage	~1,400	645/725	645/725
Hedge Instrument	SPY	SPY	Sector ETF

Ledoit-Wolf shrinkage (10 components, 47.4% var explained) improved PCA Sharpe from 0.52 to 0.61. ETF strategy with sector hedging shows comparable drawdown profile.